

Cs302

Digital Logic Design

Most Repeated Subjective Midterm

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Cs301 Midterm All Repeated Subjective

<https://youtu.be/JVduGJr3tmA>

Short Question

Question No 1

Name the four OLMC configurations

Answer

The four OLMC configurations are

- Combination Mode with active-low output
- Combinational Mode with active-high output
- Registered Mode with active-low output
- Registered Mode with active-high output

Question No 2

Why a 2-bit comparator is called parallel comparator?

Answer

The 2-bit Comparator discussed earlier is considered to be a parallel Comparator as all the bits are compared simultaneously. External Logic has to be used to Cascade together two such comparators to form a 4-bit Comparator.

Question No 3

How can a PLD be programmed?

Answer

PLDs are programmed with the help of computer which runs the programming software. The computer is connected to a programmer socket in which the PLD is inserted for programming. PLDs can also be programmed when they are installed on a circuit board

Question No 4

How many input and output bits do a Half-Adder contain?

Answer

The Half-Adder has a 2-bit input and a 2-bit output.

Question No 5

Explain the difference between 1-to-4 Demultiplexer 2-to-4 Binary Decoder?

Answer

The circuit of the 1-to-4 Demultiplexer is similar to the 2-To-4 Binary Decoder described earlier figure 16.9. The only difference between the two is the addition of the Data Input line, which is used as enable line in the 2-to-4 Decoder circuit figure

Question No 6

Name any two modes in which PALs are programmed.

Answer

PAL devices are programmed by blowing the fuses permanently using over voltage.

Question No 7

Explain Combinational Function Devices?

Answer

Xor, Xnor, NAND, NOR are combinational function devices.

Question No 8

Decoder function?

Answer

Encoder function.

Question No 9

Provide some of the inputs for which the adjacent 1s detector circuit have active high output?

Answer

The Adjacent 1s Detector accepts 4-bit inputs.

If two adjacent 1s are detected in the input, the output is set to high.

Input combinations will be

1. 0011,
2. 0110,
3. 0111
4. 1011,
5. 1100,
6. 1101,
7. 1110 and
8. 1111

The output function is a 1.

Question No 10

Draw the Truth-Table of NOR based S-R Latch

S	R	Action
0	0	Keep State
0	1	Q=0
1	0	Q=1
1	1	Restricted Combination

Question No 11

How standard Boolean expressions can be converted into truth table format.

Answer

Standard Boolean expressions can be converted into truth table format using binary values for each term in the expression. Standard SOP or POS expressions can also be determined from a truth table.

Question No 12

Briefly state the basic principle of Repeated Multiplication-by-2 Method.

Answer

Repeated Multiplication-by-2 method allows decimal fract

Question No 13

Name the three declarations that are included in “declaration section” of the module that is created when an Input (source) file is created in ABEL.

Answer

Device declaration, pin declarations and set declarations.

Question No 14

For a two bit comparator circuit specify the inputs for which $A > B$

Answer

1. 01 00,
2. 10 00,
3. 10 01,
4. 11 00,
5. 11 01 and
6. 11 10

Question No 15

When an Input (source) file is created in ABEL a module is created which has three sections. Name These three sections.

Answer

The three sections are:

- Boolean Equations
- Truth Tables
- State Diagrams

Long Questions

Question No 1

Explain Tri-State Buffers with the help of block diagram

Answer

Tri-State Buffers

Tri-State Buffer is a NOT gate with a control line that disconnects the output from the input. When the control line is high the buffer operates like a NOT gate and when the control line is low the output is disconnected from the output and high impedance is seen at the output. Tri-state buffers are used to disconnect the outputs of devices which are connected or share a common output line.

Question No 2

Differentiate between hexadecimal and octal number system

Answer

Octal – base 8

Hexadecimal – base 16

Octal and hex are used to represent numbers instead of decimal because there is a very easy and direct way to convert from the “real” way that computers store numbers (binary) to something easier for humans to handle (fewer symbols). To translate a binary number to octal, simply group the binary digits three at a time and convert each group. For hex, group the binary digits four at a time.

Question No 3

Two CMOS 5 volt series AND gates are connected together.

Answer

The first AND gate has both its inputs connected to logic high, therefore the output of the gate is guaranteed to be logic high. The logic high voltage output of the first AND gate is assumed to be 4.6 volts well within the valid V_{OH} range of 5-4.4 volts. Assume the same noise signal (as described earlier) is added to the output signal of the first AND gate

Question No 4

One of the ABEL entry methods uses logic equations; explain it with at least a single example.

Answer

In ABEL any letter or combination of letters and numbers can be used to identify variables.

ABEL however is case sensitive, thus variable „A“ is treated separately from variable „a“.

All ABEL equations must end with „;“

Boolean expression $F = AB' + AC + (BD)'$ is written in ABEL

As $F = A \&! B \# A \& C \# ! B \& ! D;$

Question No 5

Explain Carry propagation in Parallel binary adder?

Answer

Parallel binary adder:

A binary adder circuit is described using dynamic transistor logic in which for high speed carry propagation the adder stages are grouped in pairs or larger numbers and additional dynamic logic means is provided in each group to control a single transistor connected in series in the carry propagation path over the group.

The transistors used in the specific embodiments are MOS transistors, but some or all of these could be replaced by junction FET's or bipolar transistors.

Question No 6

Explain “AND” Gate and some of its uses :

Answer

AND gates are used to combine multiple signals, if all the signals are TRUE then the output will also be TRUE. If any of the signals are FALSE, then the output will be false. ANDs aren't used as much as NAND gates; NAND gates use less components and have the advantage that they be used as an inverter.

Question No 7

Write down different situations where we need the sequential circuits.

Answer

Digital circuits that use memory elements for their operation are known as Sequential circuits. Thus Sequential circuits are implemented by combining combinational circuits with memory elements.

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